

Baseline Model for NYC Ride Demand Forecasting

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Abstract

The following is the statistics for the Baseline Model for NYC Ride Demand Forecasting using XGBoost. Besides the overall statistics, I also stated the potential problems, limitations, and important finding in this baseline model in order to improve the final model.

1 Absolute Error/Residual Summary and Finding

Table 1: Absolute Error Summary

Statistic	Absolute Error
Count	720061.000000
Mean	9.116391
Std	13.844857
Min	0.000005
25%	2.187675
50%	5.257342
75%	11.060915
Max	983.192139

Table 2: Residual Summary

Statistic	Residual
Count	720061.000000
Mean	-1.689658
Std	16.490416
Min	-367.717285
25%	-7.035645
50%	-1.693508
75%	3.017321
Max	983.192139

1.1 Finding and Thoughts

Over 50 percent of the residual are negative, which means that the model tends to overpredict the demand. The mean of the residual is -1.689658, which indicates that the model tends to overpredict the demand by 1.689658 on average. The standard deviation of the residual is 16.490416, which indicates that the model's predictions are not very accurate. The maximum absolute error is 983.192139, which indicates that the model's predictions can be very far from the actual demand. However, the interval between the 25th and 75th percentile is relatively small, which indicates that the model's predictions are relatively accurate for most of the observations. Overall, the model's predictions are not useful for making practical predictions, but they are not completely useless either. The model can be improved by using more features and by using a more complex model.

2 Demand Prediction Error By Interval

Table 3: Demand Prediction Error of Baseline Model by Demand Level

Demand Level	Observations	Avg. Demand	Avg. Predicted Demand	Avg. Residual	MAE	Total Demand	Total Abs. Error	Normalized MAE
(0, 10]	140760	5.038789	7.423070	-2.384281	3.090429	709260	4.350087×10^5	0.613328
(10, 50]	305179	27.376723	30.217375	-2.840651	6.188264	8354801	1.888528×10^6	0.226041
(50, 100]	149991	71.503144	73.726020	-2.222876	10.422685	10724828	1.563309×10^6	0.145765
(100, 200]	91639	139.016390	139.665621	-0.649231	16.476590	12739323	1.509898×10^6	0.118523
(200, 500]	30531	273.331728	265.734929	7.596800	31.886967	8345091	9.735410×10^5	0.116660
(500, ∞)	1961	634.994901	560.119652	74.875249	98.966230	1245225	1.940728×10^5	0.155854

2.1 Finding and Further Change Analysis

The demand prediction is not perfect, but it provides some positive information. The predictions for demand between 10 and 500 have relatively low normalized MAE, which means that the model is able to predict demand in this range with relatively low error.

However, the predictions for demand below 10 and above 500 have relatively high normalized MAE, which means that the model is less accurate in these

ranges. This may be due to the smaller number of observations in these ranges, which makes it more difficult for the model to learn their underlying patterns. Therefore, further analysis is needed to identify ways to improve the model’s performance in these ranges.

Moreover, the general trend of the residuals indicates that the model tends to overpredict demand when the actual demand is between 0 and 200, while it tends to underpredict demand when the actual demand is above 200. This behavior is understandable for a baseline model. In the final model, we will further analyze these residual patterns and improve both the predictor set and the mathematical formulation of the model.

3 Naive and XGBoost Baseline Model Performance and explanations

Table 4: Naive Baseline Model Performance

Metric	Value
Train Size	3,497,789
Test Size	720,061
MAE	16.1457
RMSE	33.7749

Table 5: XGBoost Baseline Model Performance

Metric	Value
MAE	9.1164
RMSE	16.5767
R^2	0.9463